Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/22/2008
PI Name:	Dinsmore, Anthony Ph.D.		
Project Title:	Gelatin of Colloidal Particles on Droplet Surfaces, Dimension Curvature, and Droplet Elasticity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Colloid science		
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Space Biology Special Category:	None		
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Zip Code:	01003-9300	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
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No. of Post Docs:	1	**No. of PhD Degrees:**	1
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (4/08)		
Key Personnel Changes/Previous PI:	In July of 2005, A. J. Levine, co-PI, moved to the Department of Chemistry and Biochemistry of the University of Callifornia Los Angeles. He holds a tenure track faculty position there and became a sub-contractor of this project.		
COI Name (Institution):	Levine, Alex J. (University of California, Los Angeles)		
Grant/Contract No.:	NNC05GA40G		
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Task Description:	The goal of this project was to measure and develop theoretical models for the behavior of small particles adsorbed at two-dimensional liquid interfaces such as the surfaces of oil droplets in water or water droplets in air. The elasticity, stability, and coalescence of liquid droplets are central to many microgravity operations and industrial processes. Droplets are commonly used in waste removal, fire suppression, heat transfer, encapsulation, coatings deposition, and mineral purification. Moreover, droplet dynamics play a key role in the boiling of liquids in microgravity. In these examples, the surface tension and surface viscoelasticity determine the droplet behavior. In particular, the behavior of liquid droplets containing small suspended particles opens a number of fundamental questions in complex fluids, as well as a number of technological challenges. Many fluids of interest – paint, coffee, blood, lubricating oil, and dusty air – contain suspended particles. Because of the fluid interfacial tension, these small particles tend to adsorb at the interface and make it viscoelastic; in many cases, these particles suppress droplet coalescence or stabilize non-spherical droplet shapes. We have studied the behavior of particles adsorbed at liquid interfaces and made new findings regarding the electrostatic and capillary interactions among them, the hydrodynamics associated with their motion, and the properties of aggregates or crystallites that they form. The results contribute to a broader understanding of random solids generally, as well as a set of problems with direct relevance to microgravity operations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of these studies are relevant for understanding the behavior of liquids handled in microgravity, where relatively large fluid droplets may be suspended in air or vacuum. The results are also applicable to fluid-handling operations on Earth because many industrial processes rely on handling of fluids and because the assembly of small particles at liquid interfaces offers a potentially useful approach to forming novel materials. The specific insights of this work include the structure and dynamics of small particles on droplet surfaces, the stiffness of a dense network of such particles, and the morphology of droplets coated by such particle networks.
	Dinsmore (PI, University of Massachusetts Amherst) and Levine (subcontractor, University of California, Los Angeles) have completed the project after being granted a no-cost extension through September of 2007. I. Hydrodynamics of particles on flat interfaces and on droplet surfaces A combination of experiments and theory have led to new insights into the hydrodynamics of particles moving on crowded droplets, which are modeled as viscous membranes. This problem is an essential part of the project's overall goal of understanding interactions and dynamics of particles on the surfaces of liquid droplets because these surfaces become crowded and hence very viscous. While this problem is also relevant in other physical circumstances (notably for cell membranes), our investigations of flow fields and hydrodynamic interactions are an original and important contribution to this project. Levine and postdoc Mark Henle extended previous membrane hydrodynamic calculations by calculating the mobility of particles trapped in a membrane or at a liquid-liquid interface having non-flat geometry. To begin, we considered particle mobilities on a spherical droplet separating an internal fluid of viscosity eta 1 from an external fluid of viscosity eta 2 . The interface is given its own separate viscosity (or more generally a viscoelastic response function) to account for the presence of other surface active agents such as small-molecule surfactants or colloidal particles trapped at the interface. An interesting feature of this problem is that the ratio of the two-dimensional viscosity to the three-dimensional viscosity provides a characteristic length scale – the Saffmann-Delbruck length L_0– associated with the hydrodynamic flow. We have developed a complete calculation of the mobility of a single particle and of extended or irregular objects at the interface (1). The particle shape and interfacial geometry play large roles in the particle mobility. The relative values of the interior and exterior viscosities also play a major role, especially when the droplet radius becomes relatively small (e.g. a few microns in size). For instance, if the interior fluid is 10 times more viscous than the exterior fluid, the mobility of a disk-shaped particle is enhanced nearly 40% when the droplet radius is reduced from infinity to L_0/10. Levine and Henle also reported a theoretical investigation of the height fluctuations of supported viscoelastic films (2). Dinsmore and student Ryan McGorty have made the first experimental measurements of the flow field on a spherical viscous layer (a droplet's surface) when a rod-shaped particle is dragged along the interface. The data show the relevance of the droplet radius (R), the length of the dragged rod (L), and the length scale that arises from the ratio of the viscosities of the interfacial and bulk fluids (L_0). In the context of the aggregation of particles on droplets, we have found two useful and novel empirical results. The first is that the effect of changing viscosities and particle sizes is approximately accounted for by a simple rescaling of all lengths. Second, the droplet's curvature suppresses long-range flows on the surface. The experimentally derived flow fields were found to be in agreement with the theoretically predicted ones, obtained without fit parameters (1). II. Particle aggregates on a droplet Interactions among interfacial particles: Measurements of the interactions among colloidal particles adsorbed at liquid interfaces is an important part of understanding the droplet properties such as elasticity and stability. Despite their key role, these interactions have been a source of considerable debate and confusion in the literature. Earlier investigations revealed a short-range repulsion arising from a dipolar charge distribution on the particles, and a long-range attraction extending to 10's of microns. Possible explanations of the long-range attraction include interfacial deformation arising from electric charges on the particle (3-5) or deformation because of contact-line pinning on the particle's surfaces (6,7). In addition, Casimir-like attraction could arise from thermal fluctuations of the interface (8), though these are generally expected to be relatively weak. Dinsmore and graduate student Chuan Zeng measured the interactions between micron-sized latex spheres and liquid micro-droplets adsorbed at water-oil or water-air interfaces. We measure interactions by tracking the thermal motions of particles and then applying the principles of statistical mechanics (9). An important feature of this method is its ability to measure femtoNewton(fN)-scale forces; for microscopic particles, fN-scale forces are relatively strong and can dictate structure and dynamics. We also set up laser tweezers to position and then release microparticles. Among our discoveries are that the pair interaction U(r) consists of a short-range repulsion and a longer-range attraction. This not only reproduces features found in some earlier reports (3,4,10) but also shows that the attractive component is a two-body effect, not solely a result of non-additive many-body forces (11). We also ruled out contact-line pinning as the origin of the attraction. A manuscript providing a comprehensive review of this topic has been published (12). Results were presented at the 2006 and 2007 March Meetings of the American Physical Society; further results will be described in forthcoming publications.

Task Progress:

The development of a mechanical modulus via rigidity percolation: Levine in collaboration with Professor Hsuan-Yi Chen of the National Central University has developed a dissipative particle dynamics simulation of the diffusion-limited cluster aggregation of particles on a spherical surface. This work so far ignores the complex hydrodynamic interactions discussed above because, in dense systems these hydrodynamic flows generate only small corrections to the dynamics. At lower particulate area densities the hydrodynamic interactions have not been fully modeled. The particles in the simulation join to form an elastic network on the surface of the spherical droplet. The elasticity of the resulting network is then obtained from the elastic energy storage of the network under various deformation modes of the spherical droplet. To date we have studied the radial mode of droplet deformations as well as volume-preserving modes. Below a certain particle coverage the network is able to slip around the droplet and thereby relax its deformation energy. At a critical coverage of particles, the network opposes the elastic deformation of the spherical surface and thereby rigidifies the droplet. Our results have lead to a new understanding of diffusion limited aggregation on spherical droplets and refined our thinking on the modification of droplet elasticity due to linked particle networks on the droplet's surface. This work is currently being prepared for publication.

Imaging the morphology of crumpled elastic shells. With undergraduate Matthew Gratale, Dinsmore has measured the shape of particle-coated fluid droplets as the interior fluid is drawn out by osmotic pressure (i.e., as the droplets shrink). When the interfacial particles jam, the interfacial area remains nearly constant and the volume shrinks. Hence wrinkles appear, then give way to large-scale buckling. The goal of these experiments is to establish the role of droplet size, particle size, interfacial modulus, and shrinkage rate on the morphology of these crumpling droplets. Confocal microscopy provides 3D images. We measured the radii of gyration of the crumpled spheres and expand the shape in spherical harmonics to quantify the changing symmetry as the crumpling proceeds. (Manuscript in preparation.) Crystallization and melting or sublimation of particles on an interface. Levine and Dinsmore have investigated phase transitions of particles having short-range attraction and confined to two dimensions (13). (This project is synergistic with a project supported by the National Science Foundation.) Levine has worked with postdoc Moumita Das and graduate student Don Blair (partially supported by NASA) to develop a theoretical understanding. Based on Brownian dynamics simulations and analytic calculations, we proposed that the abrupt change in sublimation kinetics observed (13) at a fixed crystallite size is due to the appearance of thermally excited, free disclinations in the crystal. These topological defects introduce large stresses in the crystal allowing crack propagation that results in crystallite fission. This work was presented at the 2007 March Meeting of the Amer. Phys. Soc. and is currently being prepared for publication.

III. Cohesive granular materials: A new constitutive law for liquid coated particles and fine particular aggregates

Levine and graduate student Robert Brewster examined the limiting case in which the particle volume fraction is high and the volume of liquid is minimal. We completed the first computational study of what may be termed a cohesive granular aggregate. Here the liquid volume is reduced to a thin wetting layer coating each of the particles. The particles are large enough that thermal (Brownian) motion can be neglected. In collaboration with a team from Sandia National Lab, a modification of the commonly used SNL LAMPS molecular dynamics code was developed. A key feature of these fluids is that the interparticle adhesion is induced by Laplace pressure in the intergrain contacts in the presence of the fluid. This work (14) explores the constitutive relation of such cohesive granular (nonthermal) particle aggregates. It applies not only to liquid coated particles, but also generically to fine powders where e.g. electrostatic interactions also lead to cohesion. The work led to a generalization of the well-known Bagnold constitutive law that relates the shear stress to rate of shear strain in a flowing granular medium without cohesion. From the numerical data, we also proposed a theoretical explanation for the so-called breakdown of Bagnold scaling in cohesive particulate materials based on the concept that interparticle cohesion leads to transient particle clusters in the flowing state. These clusters, if long-lived on the scale of the typical interparticle collision time, can transmit stress in the pile in an independent manner from that proposed by Bagnold (instantaneous interparticle collisions). We have completed a second manuscript reporting on our further investigations into the origins of our new constitutive law. In that work (15), we examined the numerical simulations to locate long-lived clusters and measure the full particle contact lifetime distribution as a function of the interparticle cohesive energies and the mechanical properties of the particles themselves. The principal results of this work have recently appeared in Physical Review Letters, and the complete discussion of these results will be found in Physical Review E.

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Bibliography Type:	Description: (Last Updated: 12/15/2010)
Articles in Peer-reviewed Journals	Brewster R, Grest GS, Landry JW, Levine AJ. "Plug flow and the breakdown of Bagnold scaling in cohesive granular flows." Phys Rev E Stat Nonlin Soft Matter Phys. 2005 Dec;72(6 Pt 1):061301. PMID: 16485940 , Dec-2005
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